Prathmesh Vinze

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Education

Academic Qualifications

- **Ecole Polytechnique, IP Paris** — **Palaiseau**
  Ph.D Fluid Mechanics , Collective behaviour of phoretic swimmers — *2021–2024*
- **Indian Institute of Technology Madras** — **Chennai**
  M.S Chemical Engineering , CGPA- 9.79/10 — *2019–2021*
- **Birla Institute of Technology and Science Pilani, Goa campus** — **Goa**
  B.E(Hons) Chemical Engineering , CGPA- 8.20/10 — *2015–2019*

Notable Projects

- **Self-organisation of phoretic suspensions under external forcings** — **October 2021- Present**
  During my Ph.D. at *LaDhyX, École Polytechnique* under the supervision of Prof. **Sebastien Michelin**, I investigated the collective dynamics of Janus suspensions subjected to external chemical and mechanical forcings. Using a continuum kinetic model, I analysed how shear and confinement shape the self-organisation of chemotactic phoretic particles that drift toward the solutes released by their neighbours. This revealed three distinct dynamical regimes and showed how the emergent organisation strongly governs suspension rheology. Building on these insights, I demonstrated that chemically active wall patches can steer collective behaviour through external solute fields. The first part of this work has been published in **Physical Review Fluids**, with a second manuscript on control of phoretic suspensions using catalytic patches currently under preparation.
- **Effect of solute advection on the swimming velocity of a Janus particle** — **Feb 2021-Aug 2021**
  In collaboration with Prof. **S. Pushpavanam** during my graduate studies, I developed a theoretical framework to quantify weak advective effects on the swimming velocity of active particles exposed to external concentration gradients. Using the Péclet number as a perturbation parameter, I applied singular perturbation techniques and matched asymptotic expansions to evaluate the concentration field up to $O(Pe)$. By combining these results with the Lorentz reciprocal theorem, I obtained an analytical expression for the translational velocity valid up to $O(Pe)$. The findings were published in **Physical Review Fluids**.
- **Dynamics of an artificial swimmer in external concentration gradient** — **Mar 2020-Feb2021**
  This project, carried out under the supervision of Prof. **S. Pushpavanam**, focused on the theoretical description of Janus particle motion in imposed solute gradients. I showed that, in addition to its self-generated solute field, an external gradient induces extra slip velocities along the particle surface, leading to reorientation and drift. The relative strength of these contributions was captured through a non-dimensional activity number, and analytical expressions for both translational and rotational velocities were derived using the Lorentz reciprocal theorem. This work appeared in **Physics of Fluids**.

Peer-Reviewed Publication

- **Prathmesh M. Vinze**, Akash Choudhary and S.Pushpavanam, Motion of an active particle in a linear concentration gradient,Physics of Fluids 33, 032011 (2021) (Click here to access the paper)
- **Prathmesh M. Vinze**, S.Pushpavanam, Effect of weak solute advection on a chemically active particle under the influence of an external concentration gradient,Physical Review Fluids, Vol 6 Issue 12, 124201 (2021)(Click here to access the paper)
- **Prathmesh Vinze**, S. Michelin, Self-organisation and rheology of phoretic suspensions in confined shear flow, Physical Review Fluids, Vol 9 Issue 1, 2024 (Click here to access the paper)
- **Prathmesh Vinze**, S. Michelin, Controlling phoretic suspensions using wall catalytic patch (*under preperation*)

Research Interests and Work Experience

- My research interest lies in the mathematical modelling of transport processes, particularly with flows involving low Reynolds numbers. As part of my PhD, I modelled the collective dynamics of an autophoretic suspension under external mechanical and chemical forcings.
- During my time as a graduate student, I had the opportunity to work as a teaching assistant for the following courses:
  1. **Mass Transfer and Reaction Engineering Laboratory** (Jan 2020-May 2020)
  2. **Process Modelling and Simulation** (Aug 2020-December 2020)
  3. **Advanced Physics Lab III** (September 2022-2024)
  4. **Discovery Labs(Mechanics)** (September 2022-2024)

Selected Conferences and Seminars

- **Motion of an active particle in linear concentration gradients** at 73rd annual meeting of APS Division of Fluid Dynamics (Virtual) on 23rd November 2020
- **Self Organisation of phoretic suspensions in shear flow and confinement** at 14th edition of European Fluid Mechanics Conference from 12th-16th September 2022.
- **Self Organisation and rheology of phoretic suspensions in confined shear flows** at a workshop on Hydrodynamics at small scales: from soft matter to bioengineering, 14th June-16th June 2023.
- **Rheology of phoretic suspensions in shear flows** at 10th **B**ifurcations and **I**nstabilities in **F**luid **D**ynamics Conference 2024, 24th-28th June 2024.

References

1. Sebastien Michelin, Professor, *Ecole Polytechnique*, IP Paris, electronic address : sebastien.michelin@polytechnique.edu
2. Ludovic Bellon, CNRS Research director, ENS Lyon, electronic address : ludovic.bellon@polytechnique.edu
3. Akash Choudhary, Assistant Professor, *Indian Institute of Technology Kanpur*, electronic address : achoudhary@iitk.ac.in